

Isaac Mubiru

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PROFESSIONAL SUMMARY

Backend engineer with 4+ years of experience in cloud infrastructure, database migration, and distributed systems. Led zero-downtime migration of 20+ production applications from Parse/MongoDB to Firebase/GCP, implementing real-time data synchronization pipelines using Google Pub/Sub. Expertise in performance optimization, cloud operations (AWS, GCP), and building scalable backend services. Strong scripting proficiency in Node.js, Python, and JavaScript with experience in technical documentation and team enablement.

CORE COMPETENCIES

Database Migration

Cloud Infrastructure (GCP, AWS)

Performance Optimization

Node.js / Python / JavaScript

Real-time Data Pipelines

Technical Documentation

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer • US Remote

- Led end-to-end **database migration** of 20+ applications from Parse/MongoDB to Firebase/GCP with zero downtime, reporting directly to CTO
- Implemented real-time two-way sync between MongoDB and Firestore using Node.js and Google Pub/Sub for **message processing**
- Built 3D visualization feature for customers using video overlay and ffmpeg library, increasing buyer conversion
- Migrated file storage from Amazon S3 to GCS via **Python migration script**
- Saved company \$5,000/month by migrating all services off **AWS** to GCP

MTailor

Mar 2022 – Jul 2022

Contractor • US Remote

- Built **migration script** to read data from MongoDB and write to Firestore with complex processing logic
- Wrote **technical documentation** for engineers on new Firebase SDKs (Node.js and Python)
- Trained Ops team on Firebase Dashboard and backup procedures

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer • Singapore Remote

- Led **optimization** of PWA delivering real-time stock market prices from Firebase backend
- Built ad-hoc jobs to query over 2 million Firestore records for **performance**-critical price updates from Morningstar APIs
- Extended backend APIs using Firebase Cloud Functions and Express.js

CodeBits

Jan 2020 – Jul 2021

Software Engineer • Uganda Remote

- Led team of 4 developers building legal tech systems for NGOs in Uganda
- Architected microservices backend using Apache Kafka, Docker, and **Kubernetes**
- Built USSD service for legal aid providers communicating with Node.js backend via gRPC

PROJECTS

Parse → Firebase Migration Pipeline

Zero-downtime migration system with real-time MongoDB ↔ Firestore sync supporting parallel production traffic

Node.js, Google Pub/Sub, Firebase, MongoDB

Healthcare WhatsApp Chatbot

Integrated OpenAI + Google Vision for prescription OCR via WhatsApp Business API

Node.js, OpenAI, Google Vision, WhatsApp API

Marketplace Recommendation Engine

Collaborative + content-based filtering engine for service provider matching

MongoDB, Node.js

EDUCATION

Bachelor of Science in Computer Engineering — [Makerere University](#)

2018 – 2022

SKILLS

Languages: JavaScript, Python, Go, Rust **Backend:** Node.js, Express.js, Firebase Cloud Functions, gRPC **Databases:** MongoDB, Firestore, SQL

Cloud: GCP (Pub/Sub, GCS, Cloud Functions), AWS (S3, EC2, EBS) **DevOps:** Docker, Kubernetes, Apache Kafka